

Input

Active Soil Health
Telemetry

Regional Weather
and Climate Data

DA Region 10
Soil Zoning Maps

Historical Empirical
Soil Records

Process

Data Preprocessing
and Cleaning

Data Integration and
Contextualization

K Means Synthetic
Data Augmentation

Feature Engineering
Interaction Variables

Optuna Optimized LightGBM
Classification Analysis

Output

Actionable Soil
Health Insights

Zone Specific Primary and
Alternative Crop
Recommendations

Crop Specific Fertilizer
Application Strategies

Performance Metrics
and Visual Reports

User Interface
System Outputs